MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, LOW-POWER

TYPE 2N428

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification covers the detail requirements for a PNP, germanium, low-power switching transistor.

1.2 **Physical dimensions.** See figure 1 (TO-5).

1.3 **Maximum ratings.**

P_T 1/	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_{stg}
mW	*Vdc*	*Vdc*	*Vdc*	*mAdc*	*°C*
150	-30	-12	-20	400	-65 to +100

1/ Derate linearly 2 mW/°C for $T_A > 25°$ C.

1.4 **Primary electrical characteristics.**

Limits	f_{hfb}	h_{FE}	h_{FE}	$V_{CE(sat)}$
	V_{CB} = -5 Vdc I_C = -1 mAdc	V_{CE} = -0.25 Vdc I_B = -1 mAdc	V_{CE} = -0.35 Vdc I_B = -10 mAdc	I_C = -200 mAdc I_B = -10 mAdc
	MHz			*Vdc*
Min	10	60	20	---
Max	---	250	---	-0.25

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

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SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Transistors shall be of the design, construction, and physical dimensions shown on figure 1.

3.3.1 *Terminal-lead length.* Terminal-lead length(s) other than that specified in figure 1 may be furnished when so stipulated under contract or order (see 6.2) where the devices covered herein are required directly for particular equipment-circuit installation or for automatic-assembly-technique programs. Where other lead lengths are required and provided, it shall not be construed as adversely affecting the qualified-product status of the device, or applicable JAN marking.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

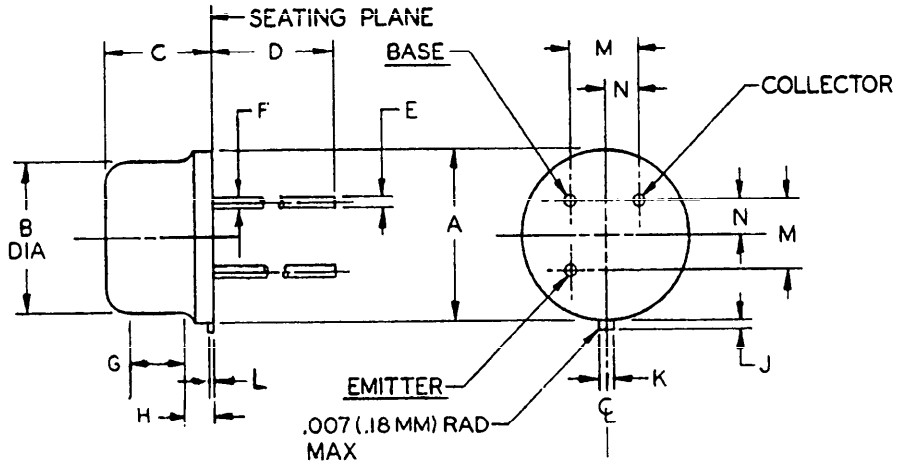
4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of group A, B, and C inspections.

4.3.1 *Group A inspection.* Group A inspection shall consist of the examinations and tests specified in table I.

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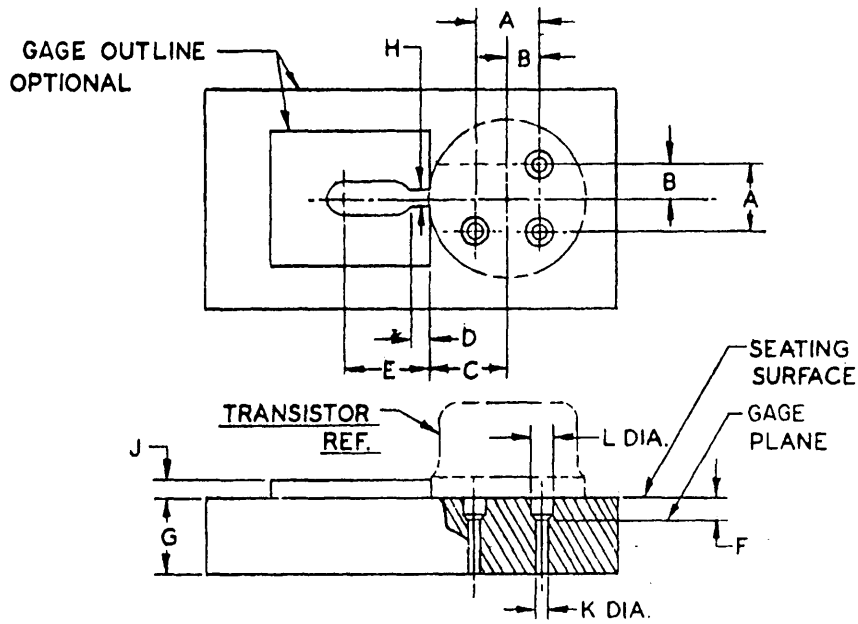


LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.335	.370	8.51	9.40	--
B	.305	.335	7.75	8.51	--
C	.240	.260	6.10	6.60	--
D	1.500	1.750	38.10	44.45	9
E	.016	.021	.41	.53	2,9
F	.016	.019	.41	.48	3,9
G	.100	---	2.54	---	4
H	---	---	---	---	5
J	.029	.045	.74	1.14	8
K	.028	.034	.71	.86	--
L	.009	.125	.23	3.18	--
M	.1414 Nom		3.59 Nom		6
N	.0707 Nom		1.90 Nom		6

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.1 mm.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. Variations on dimension B in this zone shall not exceed .010 (.25 mm).
5. Outline in this zone is not controlled.
6. When measured in a gaging plane .054 +.001, -.000 (1.37 +.03, -.00 mm) below the seating plane of the transistor maximum diameter leads shall be within .007 (.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 preferred measured method.
7. **All leads shall be electrically insulated from the case.**
8. Measured from the maximum diameter of the actual device.
9. All 3 leads.

FIGURE 1. Physical dimensions of transistor type 2N428 (TO-5).

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LTR	DIMENSIONS			
	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.1409	.1419	3.58	3.60
B	.0702	.0712	1.78	1.81
C	.182	.199	4.62	5.05
D	.009	.011	.23	.28
E	.125 Nom		3.18 Nom	
F	.054	.055	1.37	1.40
G	.372	.378	9.45	9.60
H	.0350	.0355	.89	.90
J	.150 Nom		3.81 Nom	
K	.0325	.0335	.83	.85
L	.0595	.0605	1.51	1.54

NOTES:
1. The following gaging procedure shall be used: The use of a pin straightener prior to insertion in the gage is permissible. The device being measured shall be inserted until its seating plane is .125±.010 (3.18±.25 mm) from the seating surface of the gage. A spacer may be used to obtain the .125 (3.18 mm) distance from the gage seat prior to force application. A force of 8 oz ±.05 oz shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gage.
2. The location of the tab locator, within the limits of dim C, will be determined by the tab and flange dimension of the device being checked.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 2. Gage for lead and tab location for transistor type 2N428.

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TABLE I. *Group A inspection*

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Breakdown voltage, emitter to base	3026	Bias cond. D: I_E = -25 μAdc		BV_{EBO}	-20	---	Vdc
Breakdown voltage, collector to emitter	3011	Bias cond. D: I_C = -1 μAdc		BV_{CEO}	-12	---	Vdc
Breakdown voltage, collector to base	3001	Bias cond. D: I_C = -25 μAdc		BV_{CBO}	-30	---	Vdc
Collector to base cutoff current	3036	Bias cond. D: V_{CB} = -1.5 Vdc		I_{CBO}	---	-3	μAdc
Emitter to base cutoff current	3061	Bias cond. D: V_{EB} = -1.5 Vdc		I_{EBO}	---	-3	μAdc
Subgroup 3			5				
Base emitter voltage (nonsaturated)	3066	Test cond. B: V_{CE} = -0.25 Vdc; I_B = -1 mAdc		V_{BE}	---	0.45	Vdc
Base emitter voltage (nonsaturated)	3066	Test cond. B: V_{CE} = -0.35 Vdc; I_B = -10 mAdc		V_{BE}	---	-0.8	Vdc
Collector to emitter voltage (saturated)	3071	I_C = -200 mAdc; I_B = -10 mAdc		$V_{CE}(sat)$	---	-0.25	Vdc
Forward-current transfer ratio	3076	V_{CE} = -0.25 Vdc; I_B = -1 mAdc		h_{FE}	60	250	---
Forward-current transfer ratio	3076	V_{CE} = -0.35 Vdc; I_B = -10 mAdc		h_{FE}	20	---	---
Small-signal short-circuit forward-current transfer-ratio cutoff frequency	3301	V_{CB} = -5 Vdc; I_C = -1 mAdc		f_{hfb}	10	---	MHz
Base-spreading resistance	---	V_{CB} = -5 Vdc; I_C = -1 mAdc; f = 2 MHz (see figure 3)		r_b	---	160	ohms
Open-circuit output capacitance	3236	V_{CB} = -5 Vdc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz		C_{obo}	---	20	pf
Subgroup 4			10				
Switching parameters:							
Turn-on time	3251	Test cond. A; R_B = 6,900 ohms (see figure 4)		t_{on}	---	.85	μsec
Turn-off time	3251	Test cond. A; R_B = 6,900 ohms (see figure 4)		t_{off}	---	1.10	μsec
Subgroup 5			10				
High-temperature operation:		T_A = +70°C					
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -1.5 Vdc		I_{CBO}	---	-70	μAdc
Forward-current transfer ratio	3076	V_{CE} = -0.25 Vdc; I_B = -1 mAdc		h_{FE}	50	---	---

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TABLE I. *Group A inspection - Continued*

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 5 - Continued							
Low-temperature operation:		T_A = -55°C					
Forward-current transfer ratio	3076	V_{CE} = -0.25 Vdc; I_B = -1 mAdc		h_{FE}	45	---	---

TABLE II. *Group B inspection*

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Physical dimensions	2066	(See figure 1)		---	---	---	---
Subgroup 2			15				
Solderability	2026			---	---	--	---
Thermal shock (temperature cycling)	1051	Test cond. B except T (max) = +100°C		---	---	---	---
Thermal shock (glass strain)	1056	Test cond. A		---	---	---	---
Moisture resistance	1021			---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -30 Vdc		I_{CBO}	---	-25	µAdc
Emitter to base cutoff current	3061	Bias cond. D; V_{EB} = -20 Vdc		I_{EBO}	---	-25	µAdc
Forward-current transfer ratio	3076	V_{CE} = -0.25 Vdc; I_B = -1 mAdc		h_{FE}	60	250	--
Subgroup 3			20				
Shock	2016	Nonoperating; 500 G, 1 msec; 5 blows in each orientation: X_1, Y_1, Y_2 and Z_1		---	---	---	---
Vibration, variable frequency	2056	10 G		---	---	---	--
Constant acceleration	2006	20,000 G; in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 4			20				
Terminal strength (lead fatigue)	2036	Test cond. E		---	--	---	---
End points:							
Seal (leak-rate)	1071	Test cond. G or H for fine leaks; test cond. A, C, D or F for gross leaks		---	---	$1x10^{-7}$	atm cc/sec

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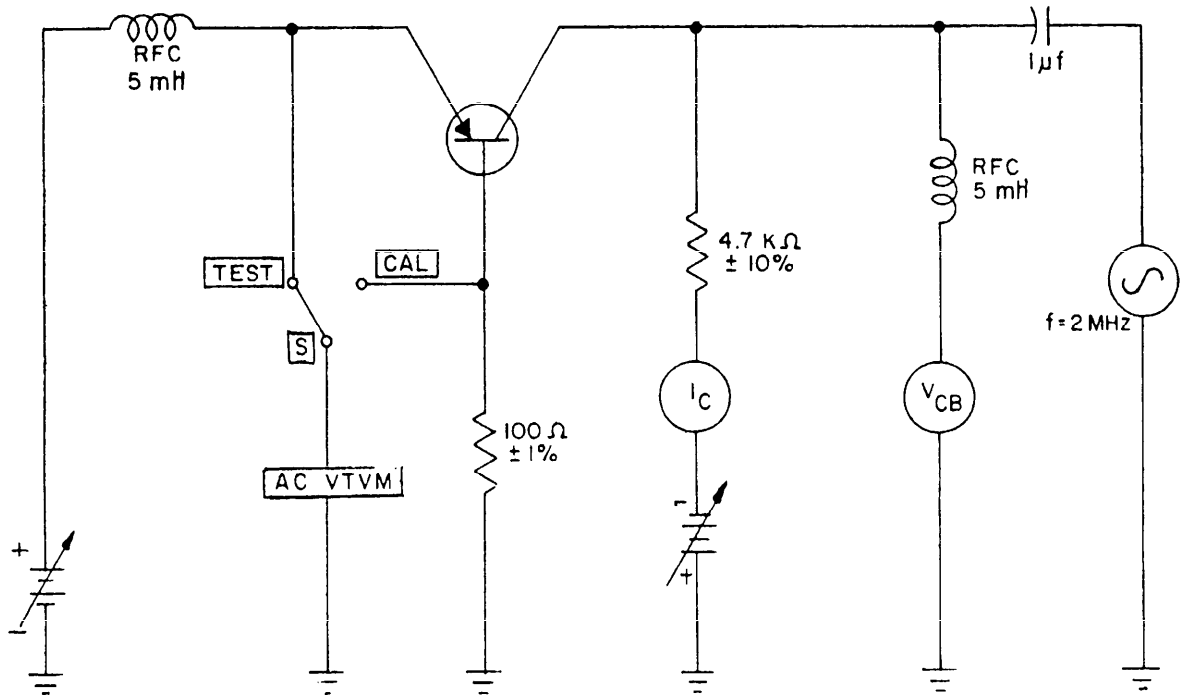
TABLE II. *Group B inspection - Continued*

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 5			20				
Salt atmosphere (corrosion)	1041			---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 6			5				
High-temperature life (nonoperating)	1032	T_{stg} = +100°C; time = 340 hours (see 4.3.4)		---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -30 Vdc		I_{CBO}	---	-40	µAdc
Emitter to base cutoff current	3061	Bias cond. D; V_{EB} = -20 Vdc		I_{EBO}	---	-40	µAdc
Forward-current transfer ratio	3076	V_{CE} = -0.25 Vdc; I_B = -1 mAdc		h_{FE}	42	300	---
Subgroup 7			5				
Steady-state operation life	1027	T_A = +25°C; V_{CB} = -15 Vdc; P_T = 150 mW; time = 340 hours (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6)							

TABLE III. *Group C inspection*

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			λ = 7				
High-temperature life (nonoperating)	1031	T_{stg} = +100°C (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6 of group B)							
Subgroup 2			λ = 7				
Steady-state operation life	1026	T_A = +25°C; V_{CB} = -15 Vdc; P_T = 150 mW (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6 of group B)							

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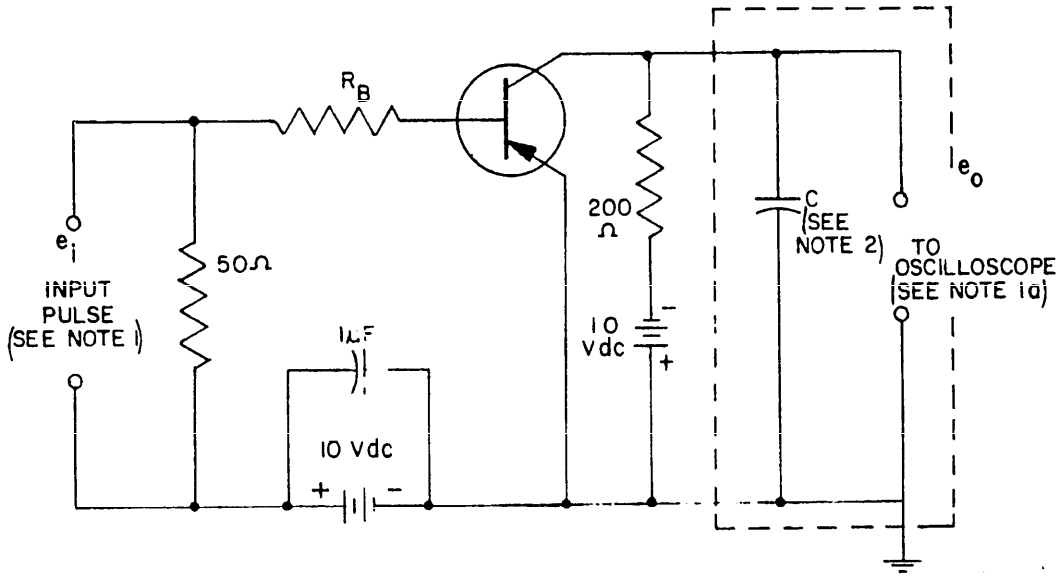
NOTES:
1. Test computation shall be based upon the following formula:

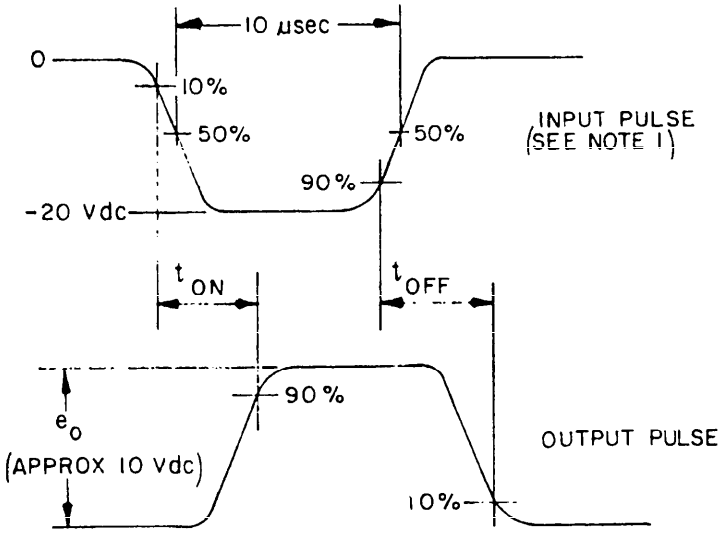
$$r_b = 100 \times \frac{\text{Reading of AC VTVM with switch S in test position}}{\text{Reading of AC VTVM with switch S in CAL. position}} - 100$$

2. An equivalent test circuit may be used.

FIGURE 3. Base spreading resistance (r_b) test circuit.

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VOLTAGE WAVEFORMS

NOTES:
1. Input pulse:
   a. The rise and fall times of the input pulse and the oscilloscope used to view the output voltage should be less than 0.03 μsec.
   b. Repetition rate = 500 Hz.
2. Oscilloscope and wiring capacitance = 15 pf.
3. An equivalent test circuit may be used.

FIGURE 4. Test circuit and waveforms for measuring t_{on} and t_{off} times.

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4.3.2 *Group B inspection.* Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 *Group C inspection.* Group C inspection shall consist of the examinations and tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every 6 months during production.

4.3.4 *Group B and group C life-test samples.* Samples that have been subjected to group B, 340-hours life-test, may be continued on test to 1,000 hours in order to satisfy group C life-test requirements. These samples shall be predesignated, and shall remain subjected to the group C 1,000-hour acceptance evaluation after they have passed the group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1,000 hours shall be computed for 1,000-hour acceptance criteria.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III.

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, section 5.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data. Procurement documents should specify the following: Terminal-lead length if other than specified in figure 1 (see 3.3.1).

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU, MI
Navy - SH
Air Force - 11, 17
DSA - ES

User activities:
Army - SM
Navy - CG, MC, OS, AS
Air Force - 13, 15, 19, 70, 80

Preparing activity:
Army - EL

Agent:
DSA - ES

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specification to the extent specified herein. Unless otherwise specified, the issues of these

SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-19500/44D, SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, LOW-POWER TYPE 2N428

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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